



Window of Sensitivity for the Estrogenic Effects of Ethinylestradiol in Early Life-Stages of Fathead Minnow, *Pimephales promelas*

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Abstract. Sexual differentiation in fish occurs after hatching during early life-stages and is believed to be a time when the gonad has a heightened sensitivity to disruption by chemicals that mimic hormones. In this study fathead minnows (*Pimephales promelas*) were exposed to an environmentally relevant concentration of ethinylestradiol (EE2) for short intervals in fish early life-stages and vitellogenic and gonadal responses were measured at 30 and 100 dph (sexual maturity), respectively. All EE2 exposure regimes induced vitellogenin (VTG) synthesis and disruption in duct development (a feminization) in males, with a window of enhanced sensitivity between 10 and 15 dph (where 60% of the males had feminized ducts). There was an altered pattern in sex cell development in males (inhibition of spermatogenesis) in the solvent controls (ethanol 0.1 ml/l) and all EE2 treatments when compared with the dilution water controls. Furthermore, fewer spermatozoa were observed in the testis of males exposed to EE2 from 15 to 20 dph and fertilized eggs (<24 h post-fertilization) -20 dph, compared with both the solvent and dilution water controls. These data show that short exposures of embryos/very early life-stage fathead minnows to an environmentally relevant concentration of EE2 lead to alterations in gonadal development that potentially have reproductive consequences and thus population level effects.

Keywords: ethinylestradiol; gonadal ducts; vitellogenin; fish; endocrine disruption

Introduction

It has now well been established that there are a wide variety of chemicals (both natural and man-made) discharged into the aquatic environment that mimic hormones and in turn may cause disruption in

development and/or reproduction in wildlife (Tyler et al., 1998). The most widely known group of hormone mimics are estrogenic in nature and these are major contaminants in effluents from sewage treatment works (STWs) that discharge into rivers in the United Kingdom and Europe. Furthermore, the estrogenic activity in UK rivers may persist for considerable distances downstream from the effluent discharges (as shown by vitellogenic responses and an inhibition of testis growth in caged fish (Harries et al., 1996)). Evidence from wild fish populations

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(both roach, *Rutilus rutilus*, Jobling et al., 1998 and gudgeon, *Gobio gobio*, van Aerle et al., 2001) living in UK rivers has shown that exposure to estrogenic effluent causes feminization of males, resulting in disruption in the development of the gonadal duct and/or, formation of oocytes in the 'testis' (Jobling et al., 1998; Nolan et al., 2001). Furthermore, recent data suggest that intersex roach are compromised in their reproductive capabilities (Jobling et al., 2002), which could have detrimental effects to the fishery.

Sexual differentiation in fish (the physical translation of the genetic sex) may occur within a few weeks of hatching, as occurs in salmonids (Nakamura, 1984), or, alternatively, it may occur much later in development, even a few months after hatching, as occurs in some cyprinid fish (Gimeno et al., 1998). Steroidal estrogens play key roles in the process of sexual differentiation and development (Conover and Kynard, 1981; Patino et al., 1996; Strussmann et al., 1996) and it is well established that exposure of fish embryos to high concentrations of steroidal estrogens can induce gonadal feminization, even in gonochoristic species (Yamamoto, 1953; Pandian and Sheela, 1995). In the medaka (*Oryzias latipes*) (Shibata and Hamaguchi, 1988) and carp (*Cyprinus carpio*) (Gimeno et al., 1998), germ cells maintain a bipotential even after gonadal differentiation and exposure to high concentrations of estrogens can also result in feminization in sexually mature adult males. The plasticity of gonadal development in fish, which contrasts with the more stable patterns found in higher vertebrates, may make fish more susceptible to endocrine-disrupting chemicals that affect reproductive development. Laboratory studies have shown that exposure of carp and medaka during early life-stages to chemicals that mimic estrogens, including the alkylphenols, octylphenol (Gray and Metcalfe, 1999), p-nonylphenol (Gray and Metcalfe, 1997), and the pesticides, beta-hexachlorocyclohexane (Wester and Canton, 1986) and methoxychlor (Nimrod and Benson, 1998) can also result in disruption in gonad development, including disruption in duct development and/or the presence of oocytes in the testis of males (Jobling et al., 1998; Rodgers-Gray et al., 2001). In most of these studies on steroidal estrogens and their mimics, however, the effects seen only occurred at concentrations far in excess of those found in the ambient environment.

Laboratory studies have shown that the concentrations of estradiol-17 β (E2; up to 80 ng/l) and estrone

(E1, up to 220 ng/l) (Desbrow et al., 1998; Rodgers-Gray et al., 2000) in treated sewage effluents entering UK rivers are sufficient to induce the vitellogenic effects observed in caged fish by Harries et al. (1996) (Routledge et al., 1998; Thorpe et al., 2000). The primary source of these natural estrogens in the STW effluents is likely to be derived from women. Although ethinylestradiol (EE2), the main ingredient of the contraceptive pill, is generally present in UK effluents at concentrations considerably lower than the natural steroidal estrogens (ranging from below the detection limit of the assay (<0.5 ng/l) up to 7 ng/l (Desbrow et al., 1998), it is a very potent estrogen indeed and can induce a wide range of adverse health effects ranging from disruption of reproduction to altered blood cell formation and function (Schwaiger et al., 2000). Furthermore, in some countries EE2 has been measured in effluents at concentrations up to 50 ng/l (Aherne and Briggs, 1989; Shore, 1993). Recent studies have shown that exposure of juvenile male medaka to 100 ng EE2/l for 2 months resulted in complete sex reversal of the population (Scholz and Gutzeit, 2000) and exposure of fathead minnow to only 4 ng EE2/l, from embryos to 172 dph, resulted in the failure of male fish to develop secondary sexual characteristics (Lange et al., 2001). These findings, together with the knowledge that exogenous steroidal estrogens may concentrate in fish (in the bile) by up to 10,000-fold (Larsson et al., 1999) emphasize the need to develop an understanding of the biological effects of environmentally relevant concentrations of EE2 in fish.

The fathead minnow, *Pimephales promelas*, a member of the cyprinid family, is one of the most widely used fish species in ecotoxicology, and its biology is well suited for laboratory-based studies. Studies on the effects of environmental estrogens on reproductive development in the fathead minnow have been facilitated by the development of assays to quantify VTG (a biomarker for estrogen exposure) (Parks et al., 1999; Tyler et al., 1999) and recent documentation of the ontogeny of gonadal development in this species (van Aerle et al., 2000). In this study fathead minnow were exposed to an environmentally relevant concentration of EE2 during early life-stages (that were within the period of sexual differentiation) and vitellogenic and gonadal responses were subsequently quantified in juvenile fish and mature adults, respectively. A primary objective of this work was to investigate windows

of sensitivity in fish early life-stages to the estrogenic effects of EE2.

Materials and methods

Embryo/fish maintenance and culture

Fathead minnow (*Pimephales promelas*) were obtained from breeding stocks maintained at AstraZeneca's Brixham Environmental Laboratory, UK. Fish were fed rotifers for the first 4 dph, artemia during the period of 5–15 dph, a combination of artemia and pellets (Promin Ltd., Hampshire, UK) during 16–56 dph, and a combination of brine shrimp and pellets during 57–100 dph. Feeding took place at a regularly adjusted rate to provide a daily ration of approximately 2% of body weight. De-chlorinated tap water (5 µm filtered) was used as dilution water to culture all embryos and fish. The parameters monitored within the test tanks throughout the study were: dissolved oxygen ranging from 4.1 to 8.2 mg/l, pH ranging from 6.87 to 7.82, and temperatures ranging from 23.9 °C to 26.3 °C. The photoperiod

was maintained at 16 h light: 8 h dark throughout, incorporating 20-min dawn: dusk transition periods.

Experimental protocol and sampling

Newly fertilized (<24 h old) fathead minnow embryos were divided into five treatment groups (100 embryos/group). In addition there were two control groups (100 embryos/group), one dilution water and one solvent vehicle control (ethanol – final concentration 0.1 ml/l). For each of the treatment and control groups the fertilized eggs were divided equally into four replicate tanks. All treatment groups received EE2 at a nominal dose of 10 ng/l, but for different exposure periods. The exposure regimes were: fertilized eggs through embryo development to 5, 5–10, 10–15, 15–20 dph, and fertilized eggs through embryo development to 20 (see Fig. 1 for experimental design). A continuous flow system was employed. A stock solution of 17 α -ethinylestradiol (100 µg/l) was prepared in absolute ethanol and dosed to the tanks using a Watson-Marlow 202U/A syringe pump equipped with 0.63 mm diameter silicone rubber tubing. The stock solution and dilution water

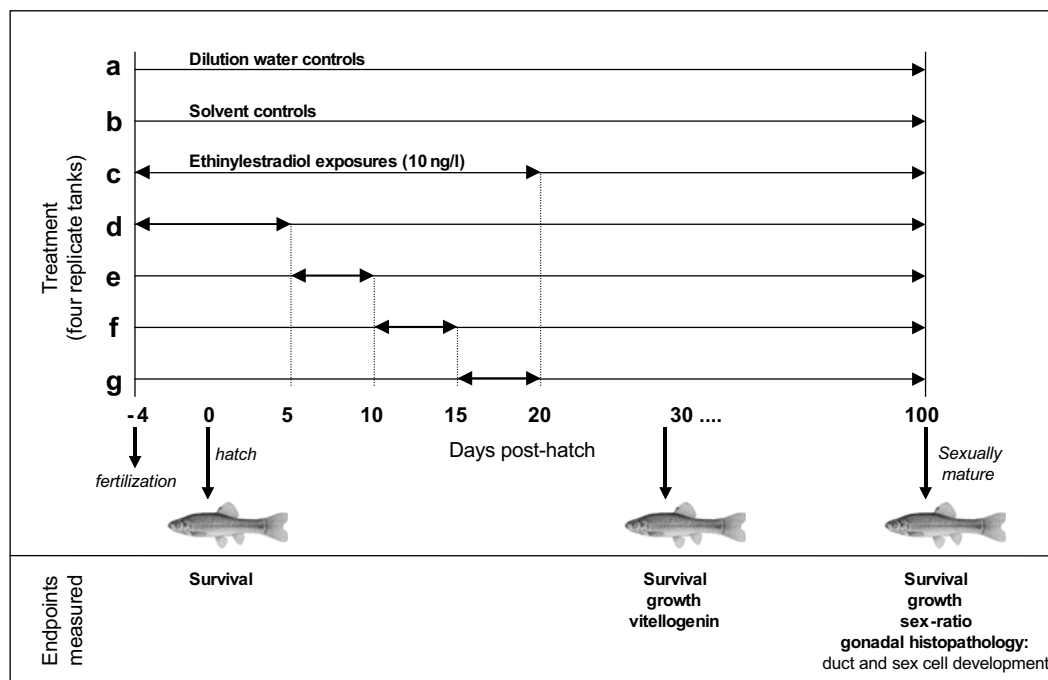


Figure 1. Experimental design for exposures of early life-stage fathead minnows to ethinylestradiol. For each control and treatment group there were four replicate tanks. End points measured were: day 0—hatchability, 30 dph—survival, growth and whole body vitellogenin 100 dph—survival, growth, sex ratio, and gonadal histopathology.

were delivered to a glass-mixing vessel prior to delivery to the test vessels at a flow rate of 250 ml/min. These all-glass vessels measured $305 \times 205 \times 210$ mm ($L \times W \times D$) and had a working volume of 9.5 l. After treatment with EE2 for the required exposure period, fish were then transferred to clean water until they were sampled at 30 and 100 dph.

The percentage embryo mortality was monitored, and subsequently any larval mortalities or developmental abnormalities were recorded on a daily basis. At 30 dph, 30 fish were randomly taken from the four replicate tanks for each treatment, sacrificed, their weights and fork-lengths measured, and snap frozen in liquid nitrogen, until required for measurement of VTG.

At 100 dph, 30 fish were randomly taken from the four replicate tanks for each treatment group for histological analysis of the gonad. Lengths and weights were taken and, where possible, the sex recorded (determined by their secondary sexual characteristics). The gonads were fixed for histological processing.

Measurement of vitellogenin

Body homogenates for measurement of VTG were prepared by homogenizing the whole bodies of individual fathead minnows in ice-cold sodium phosphate buffer (0.01 M sodium phosphate, 0.9% NaCl, 0.05% Tween 20, pH 7.4) in a 1:1 ratio wet weight:buffer volume, using a glass hand homogenizer. The homogenates were centrifuged at $3,000 \times g$ for 10 min and the supernatant withdrawn and deep frozen at -20°C , until required. Vitellogenin was measured using a carp VTG ELISA validated for use in the fathead minnow according to Tyler et al. (1999). Concentrations of VTG were expressed as ng VTG/ml.

Histological processing and analysis

Fish (100 dph) were fixed *in toto* in Bouin's fixative for 24 h and transferred into 70% industrial methylated spirits (IMS) for storage before histological processing. Whole body transverse sections (5–8 mm thick) were carefully taken from the mid-body region, between the operculum and the dorsal fin, to encompass the whole length of the gonad from each fish collected, dehydrated in IMS up to 100%, and embedded in paraffin wax (Merck Eurolab Ltd.,

Poole, Dorset, UK), using a Shandon tissue processor (Model: Citadel 2000). Serial sections were cut at 3–5 μm from the anterior end of the gonad through to the posterior end, floated in a water bath and collected onto glass slides. Sections were stained with Harris' haematoxylin and eosin (Merck Eurolab Ltd., Poole, Dorset, UK), treated with DPX mountant (Merck Eurolab Ltd., Poole, Dorset, UK) and analysed by light microscopy.

Gonad sections were examined for the presence of a female duct (ovarian cavity) or a male sperm duct and/or any abnormalities in the reproductive tract and/or germ cells. In the testis in 'males', the effects of EE2 on testicular sex cell development were also investigated. This was done by counting the number of the different cell types in four sections (randomly taken from different parts of the gonad) from each testis. The cell types counted were: spermatogonia (both spermatogonia stages, A and B, were scored together), spermatocytes, spermatids and spermatozoa. For each section two areas of 1 mm^2 were scored using a 10×10 (0.1 mm) grid and the proportions of the different cell types determined by summing the number underlying the grid intersections.

Statistical analysis

Differences for survival, weight, length (at both 30 and 100 dph), and VTG concentrations in whole body homogenates (30 dph) were analysed using analysis of variance (ANOVA). Data were not normally distributed and a Kruskal–Wallis One Way ANOVA on Ranks was performed. In cases of significant differences between the groups, post hoc tests according to Dunn's method were used to identify groups that differed significantly from the control groups. Differences in sex ratio and in the proportion of the different male sex cell types between the treatment groups and the control groups were analysed by Chi-square analysis.

Results

Survival, growth and sex ratio

There were no effects of the short-term exposures to EE2 on either hatchability (range between 92% and 100% throughout), or survival to either 30 or 100 dph (range between 92% and 99%). There was no

indication of any gross developmental abnormalities due to EE2 exposure. At 30 dph the mean length and weight of the control fish (dilution water) were 24.4 ± 0.3 mm, and 133 ± 5 mg ($n = 47$), respectively. At this time, there were no differences in growth in fish between tanks within a treatment. At 100 dph the mean length and weight of the dilution water control males were 53.1 ± 0.9 mm, and 1386 ± 87 mg ($n = 29$), respectively, and in the females, 45.4 ± 0.6 mm and 844 ± 47 mg ($n = 27$), respectively. There were no differences in growth in males at 100 dph between any of the treatment groups. In females at 100 dph there were differences in size (length and weight) between some of the treatment groups (Fig. 2).

At 100 dph the sex ratio in the controls and in most of the EE2 treatment groups was approximately

1:1. The percentage of females in the 15–20 dph and embryo to 20 dph exposure groups were 60% and 61%, respectively, suggesting a feminizing effect in these treatment groups, but they were not statistically different from the controls ($p = 0.834$).

Vitellogenin induction

The analysis on the body content of VTG in fathead minnow at 30 dph was carried out on mixed sex individuals, as it was not possible to identify the sex of the fish macroscopically at this time. The mean concentration of VTG in the dilution water control fish was 917 ± 74 ng/ml ($n = 45$) and there was no effect of the solvent vehicle on VTG induction (972 ± 97 ng/ml, $n = 29$). Exposure to EE2 in all

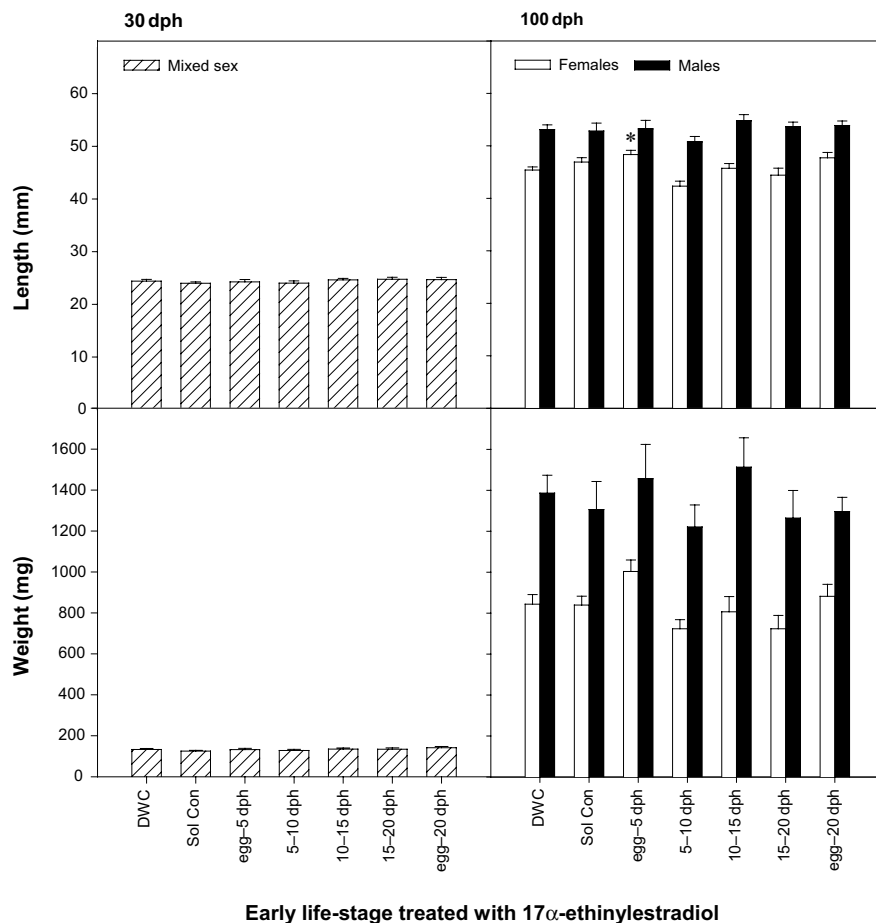


Figure 2. Length (fork) and weight of fathead minnows exposed to ethinylestradiol at 30 and 100 dph. Significant differences (pooled data from the all tanks in each treatment) from both sets of controls (dilution water and solvent) are indicated (* $p < 0.05$). Values are means $\pm$ standard error of the mean.

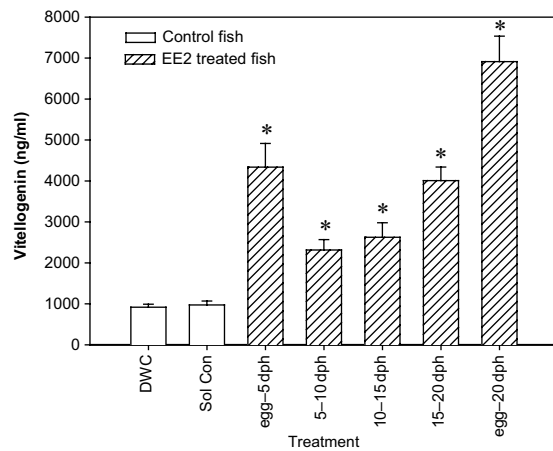


Figure 3. (a) Vitellogenin concentrations in whole body homogenates (mixed sex) in fathead minnows 30 dph after exposure to ethinylestradiol during embryo and/or larval development. Significant differences (pooled data from all tanks for each treatment) from both sets of controls (dilution water and solvent) are indicated (* $p < 0.05$). Values are means $\pm$ standard error of the mean.

treatment regimes resulted in a significant increase in whole body VTG content ($p < 0.05$, Fig. 3). VTG induction was positively correlated with longevity of exposure, and the highest induction occurred in the 24-day treatment group (fertilized eggs to 20 dph) where there was a 7-fold higher concentration of VTG than in the controls.

Disruption of duct development

In the control groups no oocytes or duct aberrations were found in any of the male fish ($n = 81$). Similarly no oocytes were seen in any males from any of the EE2 treatment groups. In all of the EE2 treatment groups, however, a proportion of males had an ovarian-like cavity. Figure 4 shows histological sections of fathead minnows at 100 dph showing – a maturing ovary with an ovarian cavity (a), a normal testis with a sperm duct (b), a testis with a complete ovarian-like cavity (c). The incidence of ovarian-like cavities in males in the different EE2 treatment varied between 8.3% and 64.3% (Fig. 5). The highest incidence of ovarian-like cavities in males (64.3% of the mature males) occurred in fish exposed to EE2 during 10–15 dph. In some of these males (and in the other EE2 treatment groups) the ovarian cavity was

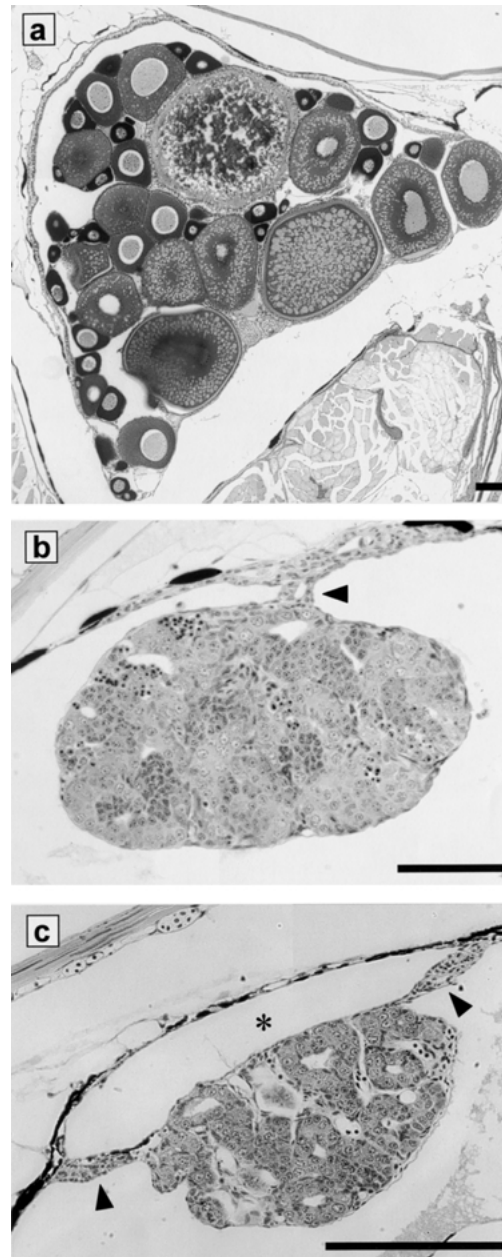


Figure 4. Histology of the gonad in control and ethinylestradiol exposed fathead minnow at 100 dph. (a) Normal female—showing ovarian cavity. (b) Normal male—showing sperm duct (arrowhead). (c) Male with fully formed ovarian-like cavity (*) attached to the peritoneal wall by duct cells (arrowhead). Bar 100 μ m.

only partially formed along the testis. On the anterior side of the gonad, a clear cavity between the gonad and the peritoneal wall was formed (Fig. 6a), whereas at the posterior edge the duct cells from one side were

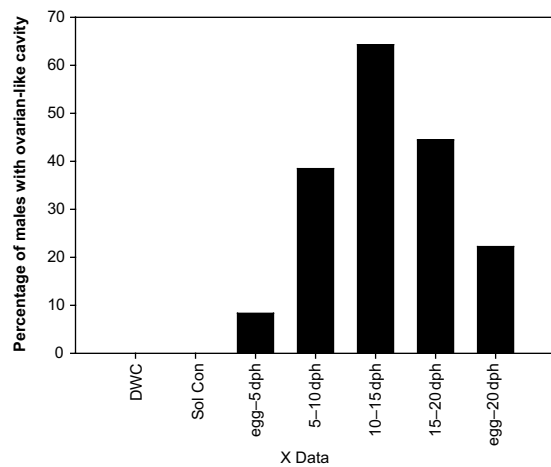


Figure 5. Incidence of fully formed ovarian-like cavities in male fathead minnows exposed to 10 ng/l ethinylestradiol during early life-stages.

separated from the gonad and appeared as well defined 'buds' of somatic cells attached to the peritoneal wall (Fig. 6b). In some fish, the duct cells stretched from one side of the gonad to the other, forming a wall, similar to the wall created during the formation of an ovarian cavity (Fig. 6c). No apparent abnormalities in duct development were seen in females in any of the treatment groups.

Sex cell development in males

The percentages of the different sex cell types in the testis of mature males at 100 dph are shown in Fig. 7. The rate of sex cell development varied between males within a treatment. The pattern of the different sex cell types in the testis, however, differed between the dilution water controls and all of the treatment groups ($p < 0.05$). There was a higher percentage of spermatozoa in the testis of the dilution water controls than in both the solvent controls and in all of the EE2 treated males. Males that had been exposed to EE2 between 15 and 20 dph and from egg 20 dph had the least developed testis in terms of their sex cell profiles; they had a lower proportion of spermatozoa than the solvent controls and the 15–20 dph exposure group also had a higher proportion of spermatogonia than both the dilution and solvent controls. No apparent abnormalities in oocyte growth and development (e.g. rate of

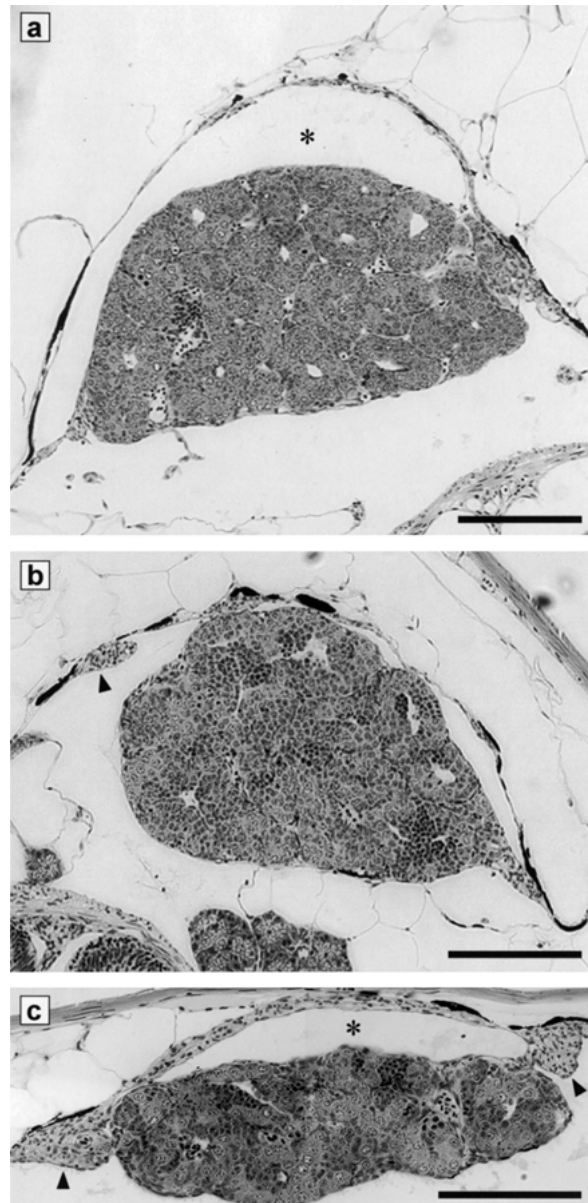


Figure 6. Histological appearance of fully and partially feminized ducts at 100 dph in male fathead minnows exposed to ethinylestradiol during early life-stages. (a) fully formed ovarian-like cavity (*), found on the anterior end of the gonad, (b) partially formed cavity, found on the more posterior end of the gonad, in which the duct cells (arrow head) are separated from the gonad and attached to the peritoneal wall, and (c) fully formed cavity (*) enclosed by duct cells (arrowhead). Bar 100 μ m.

oocyte development, size of oocytes, presence of atretic oocytes, etc.) were seen in the ovaries in females in any of the treatment groups (data not shown).

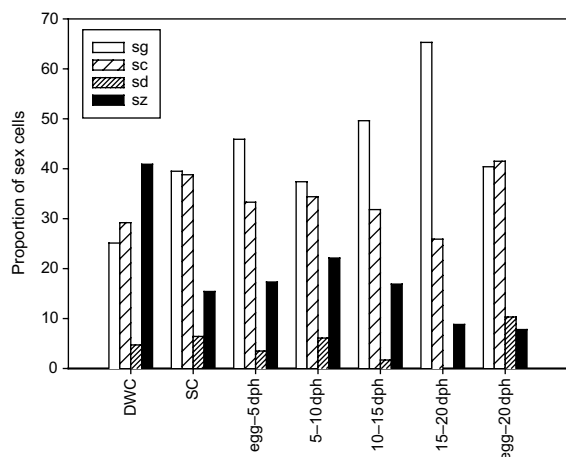


Figure 7. Percentages of different sex cell types in the testis of 100 dph fathead minnow treated with 10 ng/l ethinylestradiol during early life-stages. Spermatogonia (Sg), spermatocytes (Sc), spermatids (St) and spermatozoa (Sz).

Discussion

The data presented show that exposure of fathead minnows to an environmentally relevant concentration of EE2 for short intervals during early life-stages causes feminizing effects. Fathead minnows less than 5 days old were induced to synthesize VTG and the amount synthesized was positively correlated with the length of exposure to EE2. The amount of VTG synthesized appeared to be affected by the timing of the estrogen exposure, but there were no statistical differences in VTG concentrations between any of the EE2 treated groups with the exception of the group that was exposed to EE2 from embryos through to 20 dph, which was significantly different from the groups that were exposed from 5–10 to 10–15 dph, respectively. Interpretation of these data, however, is complicated by the fact that we do not have data on the depuration rates for VTG in the juvenile fathead minnow. At 30 dph, the lower concentrations of VTG in fish that had been exposed to EE2 earlier in development may be as a consequence of greater time period for clearance of VTG between EE2 exposure and measurement of VTG. Studies on VTG induction in fathead minnows exposed to E2 indicate a somewhat higher sensitivity to estrogen in early life-stages compared with in adults (Panter et al., 1998; Tyler et al., 1999). The magnitude of VTG induction in response to estrogen in early life-stages, however, appears to be lower than

in adult fish, although these analyses have compared concentrations of VTG in whole body homogenates (in early life-stages) with VTG concentrations in plasma (in adults). In fathead minnow early life-stages, EE2 has a potency for VTG induction approximately five times higher than E2 (comparing VTG induction in this study with that for the same life-stages exposed to E2) (Tyler et al., 1999). In a very recent study in juvenile rainbow trout (*Oncorhynchus mykiss*) EE2 was estimated to be even more potent relative to E2 in inducing VTG synthesis (approximately 25-fold) (Thorpe et al., 2001a). The estrogenic potency of EE2 in early life-stage fish reinforces the concern about the discharge of EE2 into the aquatic environment. Induction of high concentrations of VTG as a consequence of exposure to estrogens can result in kidney failure and death in fish (Herman and Kincaid, 1988). Health effects of a lower level induction of VTG as a consequence of exposure to environmental estrogens are not known. Fish early life-stages, however, constitute a period in the life of the organism when the energy budget is critically balanced and any deviation of resource away from growth and development to fuel VTG synthesis is likely to be detrimental for survival in their natural environment. The ability of a short term exposure to EE2 to induce VTG synthesis in fish early life-stages has further significance when it is realized that wild fish living in UK rivers are exposed continually to estrogenic mixtures, both naturally and anthropogenic (Tyler and Routledge, 1998), some of which are known to be additive in their effect (Thorpe et al., 2001b).

The finding that very early life-stage fathead minnows are able to synthesize VTG opens the opportunity for developing an *in vivo* assay for estrogenic chemical effects in fish that are a few days old. Fish of this age are too small to obtain sufficient tissue to quantify VTG protein using the available assays (and thus the need to grow them to 30 dph in this study), but big enough to measure and quantify expression of VTG mRNA(s) using techniques such as real time RT-PCR. Quantification of VTG mRNA(s) has been successfully employed to assess estrogen exposure in adult fish (Folmar et al., 2000) and to test for estrogenic chemicals in cultures of fish hepatocytes (Flouriot et al., 1995). The utility of VTG mRNA induction to measure estrogen exposure in fathead minnow early life-stages would be enhanced with an ability to sex juvenile fish

(e.g. with the use of a sex-specific probe) and thus separate males, which have negligible expression of VTG mRNA, from females, which synthesize VTG even at this age, albeit at low levels.

Normal gonad development occurred in all fish of both sexes in the control groups (dilution water and solvent) with no signs of disruption in sex cell development. At 100 dph, there appeared to be a bias in the sex ratio in favor of females in the groups of fish treated with EE2 either during 15–20 dph, or as embryos through to 20 dph, but they were not statistically different from the controls ($p = 0.834$). Furthermore, as no intersex fish were found in these groups (as defined by a gonad containing both male and female sex cells) it is unlikely that any sex reversal (complete feminization) occurred. Recent data indicates that (histological) sexual differentiation in female fathead minnows can occur as early as 10 dph and in males, it occurs between 60 and 90 dph (van Aerle et al., 2000), and this confirms that the EE2 exposure regimes adopted in this study occurred during the period of sexual differentiation. The lack of any oocytes in male fathead minnows exposed to 10 ng EE2/l is consistent with a study in the medaka where a minimum dose of 100 ng EE2/l was needed to induce sex cell feminization (Scholz and Gutzeit, 2000). Effective concentrations of other steroidal estrogens for gonadal feminization vary depending on the fish species, but estimates also differ for the same species and life stage between studies. For example, exposure of medaka early life-stages to 10 ng E2/l (from 5 to 8 day post hatch for 31 days) was reported to result in complete feminization of the population (Nimrod and Benson, 1998). In another study on the medaka, however, exposure of embryos to 10 ng E2/l resulted in a low incidence of phenotypic sex reversal (as determined by a color change in the d-rR strain of medaka) only, and an exposure of 1 µg/l E2 (for 20 days from fertilization to 10 dph) was required to induce gonadal feminization (Iwamatsu, 1999). In the study by Iwamatsu and colleagues, ten times the amount of E2 was required to cause sex reversal in adult medaka compared with embryos. In masu salmon (*Oncorhynchus masou*) and chum salmon (*Oncorhynchus keta*), in which morphological sex is distinguishable at 13 and 25 dph, respectively, an exposure concentration of 500 ng E2/l during early life-stages (for 2 weeks) was required to induce complete sex reversal (Nakamura, 1984).

In control fish (both dilution water and solvent) at 100 dph, all gonadal ducts were normal, in both males and females. In contrast, in all the EE2 treatment groups a proportion of male fathead minnows had an ovarian-like cavity. In many of these males the cavity was partially formed. It should be emphasized that our ability to identify a partially formed ovarian-like cavity was because our sectioning procedure encompassed the full length of the gonad. The formation of the ovarian cavity in female fathead minnow follows the same pattern as in carp and roach, where a small aggregation of somatic cells appear on the peritoneal wall adjoining the extended ridge of the gonads and eventually fuse with that part of the gonad to form a narrow cavity in the region between the gonad and the peritoneal wall (Nakamura, 1978). The presence of fully and partially formed ducts demonstrates that the process of the formation of the female-like duct in exposed males is very similar to the formation of the ovarian cavity in females. The histological sections analysed from different parts of the gonad suggest that the ovarian-like cavity is formed from the anterior end of the gonad towards the posterior end and that the duct cells grow from the peritoneal wall towards the gonad. It is likely that the 5-day windows of exposure to 10 ng/l EE2 were too short to induce the complete feminization of gonadal ducts in some fish and exposure for longer periods during sex differentiation and/or exposure to higher concentrations of EE2 would be required to do so. This is supported by results from a recent study by Zillioux and colleagues, where sheepshead minnow (*Cyprinodon variegatus* Lacépède) were exposed to various concentrations of EE2. In that study, testes of male fish exposed to 20 ng/l EE2 for 59 days (starting from 6 weeks post-fertilization) contained large numbers of primary oocytes. Higher concentrations ranging from 200 to 800 ng/l resulted in (almost) complete feminization (Zillioux et al., 2001).

The great sensitivity of the gonadal duct to disruption by estrogens parallels that seen in other fish (e.g. carp (Gimeno et al., 1997) and tilapia, *Tilapia mossambica* (Nakamura and Takahashi, 1973)). A recent study that exposed early life stage roach, *Rutilus rutilus*, to estrogenic treated sewage effluent during the period of sexual differentiation, similarly resulted in an ovarian cavity in 'males' (100% of males exposed to full strength effluent had an ovarian cavity), but did not induce oocytes (Rodgers-Gray et al., 2001). The presence of an

ovarian cavity (partial, or fully formed) in adult fathead minnow exposed to estrogens as embryos or juveniles adds further support to the suggestion that disruptions in duct development during early life are permanent (Rodgers-Gray et al., 2001). The implications of duct disruption/feminization for wild fish populations have not been determined. Recently, in the roach, however, it was established that gamete quality in intersex fish was inversely related to the degree of gonadal feminization (Jobling et al., 2002). Aromatase expression in the gonad (Scholz and Gutzeit, 2000) and alterations in male secondary sexual characteristics (Harries et al., 2000) are also more sensitive to exogenous estrogens compared with induction of oocytes in the testes.

The highest incidence of ovarian-like ducts in this study occurred in 'males' (60%) that had been exposed to EE2 during 10–15 dph. Critical windows for the feminizing effects of estrogens have been reported in other fish. As an example, incomplete feminization was observed in tilapia (*Tilapia mossambica*) exposed to EE2 (via the diet, at a rate of 50 µg/g food) for the period 11–25 dph, but exposure to the same concentration during 6–25 dph caused complete feminization (Nakamura and Takahashi, 1973). Different life-stages of salmon too, show different sensitivities to different toxicants (McKim et al., 1996). Interestingly, male fathead minnows in this study exposed to EE2 from embryos to 20 dph had a lower incidence of duct disruption than in fish exposed to EE2 for 15–20 dph. This indicates a possible de-sensitizing effect for duct disruption when fish are exposed to EE2 for periods of time that extend before the critical window for duct development/disruption. Together these data emphasize that the timing of exposure to environmental estrogens is crucial for effects on gonadal development.

It has been established in other fish species that exposure to high concentrations of androgens and estrogens during early life-stages can have permanent detrimental effects on spermatogenesis (Yamazaki, 1983; Pifferer et al., 1994). Similarly, exposure to high concentrations of environmental estrogens, such as nonylphenol (Jobling et al., 1996) and bisphenol A (Sohoni et al., 2001) may also cause an alteration in testis cell development in rainbow trout (Jobling et al., 1996). This study on the fathead minnow, shows that exposure even to relatively low levels of a steroidal estrogen for brief intervals during early development

can also affect the numbers of the different sex cells in the adult testis, causing a retardation in sex cell progression. The consequences of this, if any, on reproductive performance are presently not known. The finding that the solvent vehicle (here ethanol) caused a retardation in spermatogenesis parallels some other recent findings in female rainbow trout where the solvent methanol was found to suppress plasma concentrations of sex steroids (Harris et al., 2001) and emphasizes the need for further work regarding the use of solvents in fish toxicology studies. No apparent abnormalities in oocyte growth and development (e.g. rate of oocyte development, size of oocytes, presence of atretic oocytes, etc.) were seen in the ovaries in females in any of the EE2 treatment groups.

In summary, exposure to an environmentally relevant concentration of EE2 during early life-stages in the fathead minnow induces VTG synthesis, and causes disruption in the development of the gonadal duct and the progression of sex cell development in males emphasizing the vulnerability of early life stage fish to the endocrine disrupting effects of this synthetic steroidal estrogen.

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